

Unus Solus Possibilis Est

Corpus Closure and Uniqueness of the Admissible Interior

Greatestness, complete representation, and faithful transport

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Abstract

We construct the unique greatest admissible interior for an original task whose local qualification, warranted transformations, and required uses are fixed independently of a proposed answer, and prove its invariance under exact changes of description. There is exactly one complete maximal region in which every declared continuation remains qualified and warranted: the greatest admissible interior. We distinguish this greatest region from a least generated closure, a chosen invariant subset, a minimal complete interface, and an individual occupant. An exact criterion determines when a surjective description map carries the entire safety operator to a quotient: effective local validity (immediate qualification together with all outgoing warrants) must be constant on fibres, and locally valid representatives must have the same sets of successor classes. This criterion transports every finite-depth approximation and the greatest interior. Minimal complete interfaces are uniquely isomorphic, and faithful maps preserve and reflect interior membership under explicit successor-matching conditions. For several contracts on one carrier we construct the greatest common interior, identify exactly when it equals the intersection of their separate interiors, and characterize conservative additions of obligations. Finally, every incorrect complete classifier or proposed interior has an explicit surplus, omission, or finite unsafe-path witness. The resulting uniqueness concerns the complete admissibility structure for the same original problem. The construction supplies an explicit theorem chain for corpus applications. Conclusions about individual occupants, physical laws, or numerical values require the corresponding domain constraints, identity separation, and existence proofs.

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1 The uniqueness claim and its proof architecture

1.1 What the title asserts

Unus Solus Possibilis Est asserts the uniqueness of the complete admissibility structure for one original problem. We establish that uniqueness by constructing the greatest admissible interior and proving its invariance under exact changes of description. Fix the original candidates, the questions they must answer, the transformations under which they will be reused, and the warrants required for those transformations. Then there is one greatest region satisfying all those demands. Every sound region lies inside it; every complete sound account gives exactly it. Exact changes of representation carry that region to the corresponding region in the new representation. A proposed complete answer for those requirements is tested by the soundness, completeness, and representation conditions proved below.

The word *complete* matters. A conservative certificate may recognize only some genuinely safe cases. A laboratory may choose only one of several safe designs. Neither operation determines the full region. The theorem concerns the complete answer to the original qualification-and-reuse question, with all its declared uses retained. The original data specify the qualification and reuse obligations independently of that interior.

1.2 A familiar reading

Consider a collection of records for a repeated test. A record can satisfy its immediate checks while still allowing a declared next step that invalidates its use. The robust question is therefore not merely “does this record pass now?” but “do all the continuations for which this record is being certified preserve the required checks?” One can remove the immediately bad records, then records that lead to them, and continue. The survivors constitute the greatest admissible interior. Different laboratories can print different identifiers for the same records, but a description that merges a safe record with an unsafe one cannot give the complete original answer. Nor can selecting a single successful record establish that it was the only possible survivor.

This example separates three responsibilities. Original qualification specifies what the test requires. Closure under declared continuation specifies the strength of the certification. Representation specifies which distinctions a record system retains. None can silently replace either of the others. The worked examples in Section 9 exhibit surplus records, omitted safe records, unsafe continuations, and lost information explicitly.

1.3 The theorem chain

The main results follow a single dependence chain.

1. **Independent original requirements.** Definition 2.1 fixes the candidate space and its qualification and reuse obligations before an interior is proposed.
2. **One complete interior.** Theorem 3.1 and corollary 3.2 construct the greatest sound region and prove that it is the unique inclusion-maximal sound region. Corollary 3.3 identifies the exact additional obligation for a positive realization.
3. **Exact descent to descriptions.** Theorems 4.1 and 4.2 characterize when an interface supports the same one-step safety calculation and transports the entire greatest interior.
4. **Rigidity of complete realization.** Theorems 5.1 to 5.3 establish the unique identification of minimal complete descriptions and standing preservation and reflection across faithful maps.

5. **Joint use and extension.** Theorems 6.1, 6.2 and 6.4 construct the interior for several contracts and identify when adding obligations conserves the original interior.
6. **Conservation and exhaustion.** Theorems 7.1, 8.1 and 8.2 turn incorrect claims into witnesses. Theorem 10.1 gathers the precise result available for downstream invocation.

The first construction is the greatest-interior theorem already developed in *The Structure of Admissibility* [6]; a short self-contained proof is included to fix the object whose uniqueness is exported. The additional work here concerns its complete realization, exact quotient descent, combination of contracts, and conservation across descriptions. Together they establish the hypotheses and conclusions available for each downstream uniqueness application.

1.4 Necessary Kernel governance and the division of labour

The Kernel identifies the necessary work of Admissibility, Standing, Reference, and Irreversibility in non-degenerate determinate objecthood [4]. A determinate state, record, carrier, or endpoint does not first become a complete independent object and then optionally acquire that governance. The present task data already involve distinguishable candidates, attributable answers, identifiable transformations, and determinate incidences. Their use therefore presupposes that work. Fixing an original task specifies *which* qualification and reuse question is being answered; it does not decide whether the Kernel applies.

This necessity does not identify every domain's qualification predicate or transition relation. A bridge trial and a proof system incur the same necessary roles while answering different original questions. A domain proof must identify those questions and show that the mathematical data adequately express them. In particular, a determinate record can fail certification for a requested reuse while remaining a determinate record of that failure. Negative qualification is not the annihilation of its identity.

The upstream division of labour is precise. The Kernel supplies the necessary roles. The AMetric boundary supplies the pre-fixational restrictions on licensed content, symmetry breaking, and transmission [5]. The Structure manuscript supplies complete interfaces and the greatest safe interior. The fixed-domain exhaustion paper supplies coverage and composition for generated uses [7]. ATS supplies the factorization of variation effects and identity relative to a declared envelope [8]. This paper proves how one complete interior is uniquely represented, faithfully transported, jointly constrained, and defended against claimed alternatives. The downstream applications in Section 10 illustrate the resulting export conditions; the present proofs depend on the independently stated task data.

1.5 Mathematical neighbours and the contribution

The order-theoretic engine is classical. Tarski's fixed-point theorem gives greatest and least fixed points of monotone operators on complete lattices [1]. Here the relevant operator has a concrete universal-successor form; its elementary union proof and finite-path witnesses make the qualification claim directly inspectable. The contribution is the connected construction, transport, and exhaustion theorem for independently interpreted admissibility tasks.

The representation results belong near behavioural equivalence, quotients, and coinduction. Rutten's account of systems and coalgebra provides the wider mathematical setting for homomorphisms and bisimulation [2]. Our exact criterion is stated directly for the task's safety operator and distinguishes full one-step descent from accidental agreement of final status. The finite-path results concern safety, not eventual success; the safety/liveness distinction of Alpern

and Schneider is relevant here [3]. A task requiring eventual completion must represent that obligation separately.

The contribution is a connected dependency theorem: greatestness plus independently justified coverage, exact representation, and faithful transport, followed by witnessed exhaustion of purported complete alternatives. Classical machinery does not by itself establish that a particular physical or mathematical description meets these hypotheses. Conversely, once the hypotheses have been established, a second incompatible complete answer for that same task is excluded by proof, without an additional interpretive choice.

1.6 Five distinctions that must remain explicit

Object or comparison	What is being established
Greatest interior	The full subset of candidates robust under all declared uses.
Generated closure	The least region reachable from specified seeds; it equals the greatest interior exactly under the soundness and coverage conditions of proposition 3.7.
Minimal complete interface	The quotient retaining exactly the original answers.
Faithful transport	The standing answer preserved and reflected by a map satisfying the exact conditions.
Occupant	An individual candidate, physical system, mathematical object, or identity class within the interior.

The interior is unique, and minimal complete presentations are uniquely isomorphic over the original carrier. General faithful presentations need not be minimal or uniquely isomorphic; their transported standing answer is exact. Uniqueness of an occupant needs an additional separation result for the original identity question. A Boolean standing answer has only two values; it can nevertheless be shared by arbitrarily many distinct positive occupants. The theorem therefore identifies the unique complete structure and provides an explicit bridge to occupant uniqueness when the domain supplies identity separation and existence.

2 An original task fixed independently of the proposal

Definition 2.1 (Original qualification and reuse contract). An original contract is data

$$D = (X, L, R, A, W), \quad R \subseteq X \times L \times X, \quad A \subseteq X, \quad W \subseteq R.$$

The set X contains the candidates or incidences to be classified. The set L indexes the declared kinds of use. An edge $(x, a, y) \in R$ records an originally relevant use of x with outcome y . Membership in A expresses immediate qualification, including any required readiness conditions. Membership in W expresses the warrant required for an edge. These sets and their interpretation are fixed independently of a proposed interior or its claimed status answers.

An edge need not be physical motion. It can be reuse of a record, a proof transformation, a comparison, or a request made through an interface. Nondeterministic outcomes are represented by multiple edges. A partial operation's definedness is part of the original question whenever it matters. If a required operation must be available, the existence of a suitable outcome belongs in the readiness component of A . Otherwise a state with no edges would pass a universal continuation condition vacuously.

The universal reading is substantive: the certificate sought here covers *every declared outcome*. If an agent may choose a controller to avoid a bad outcome, that is a different contract. It may

be represented by controller-state pairs with the controller's actual outcomes as the edges, but the controller choice and its coverage cannot be erased from the specification.

Definition 2.2 (Local validity and the safety operator). For $x \in X$ and $U \subseteq X$, put

$$B = \{x \in A : \forall a, y ((x, a, y) \in R \Rightarrow (x, a, y) \in W)\}, \quad (2.1)$$

$$\text{Succ}(x) = \{y \in X : \exists a \in L (x, a, y) \in R\}, \quad (2.2)$$

$$\Phi(U) = \{x \in B : \text{Succ}(x) \subseteq U\}. \quad (2.3)$$

A set U is *sound for the contract* if $U \subseteq \Phi(U)$.

Soundness means both local validity and retention of that validity under every declared continuation. The aggregate B abbreviates an interpreted conjunction; it is not an unexplained primitive standing bit. The labels and individual warrants remain available in the original data even when the safety operator depends only on their aggregate effect. A quotient that preserves this aggregate calculation need not preserve an additional question about which label or warrant was used. Such a question must be retained in the complete original profile when it belongs to the task.

2.1 Original answers and adequate description

Let J index the original observations, comparisons, definedness queries, identity questions, and declared finite uses. Write their total answers as $p_j : X \rightarrow Y_j$ and their complete profile as

$$P : X \longrightarrow \prod_{j \in J} Y_j, \quad P(x) = (p_j(x))_{j \in J}.$$

Undefinedness can be represented by a distinguished answer in Y_j . The index set and interpretation must be justified by the original problem, not reconstructed from the distinctions a proposed encoding happens to retain. If a smaller generated language is used in practice, the claim that it covers these original questions requires a coverage proof; the compositional method in [7, Sections 3–4] supplies one route.

An encoding $h : X \rightarrow H$ is *adequate for these answers* if $P = d \circ h$ for a decoder d on $h(X)$. It is *minimal complete* if, in addition, equal complete profiles yield equal codes. Thus an adequate encoding may retain redundant information; a minimal complete one retains exactly the original answer distinctions. Neither definition presupposes agreement between competing classifications. The answers are fixed first; agreement is then proved from soundness, completeness, or exact decoding.

2.2 What the formalization must not decide by notation

Writing down D does not prove that X covers the actual domain, that R contains every originally claimed use, or that A states the intended qualification. Those are interpretation obligations. They are external to the elementary set-theoretic proof, but internal to an application of the theorem. A proposal that deletes a difficult use from R has changed the certification problem. A proposal that replaces an original equality question with a coarser Boolean one has changed the interface problem. Such changes can be worthwhile; they cannot be presented as two answers to unchanged requirements.

The same-task requirement consequently fixes questions and meaning, rather than fixing an answer. It does not contain a chosen S , a named gate, an ATS carrier, or a uniqueness premise. A proposed classification will be judged against A, W, R and the original answer profile.

2.3 Safety, incidence, and existence

The construction concerns robustness under reuse. All finite paths being safe does not assert that some desired endpoint is eventually reached. An infinite loop can satisfy every safety check and never finish. If termination or success is itself an original requirement, it must be added as an independently justified coordinate, a progress condition, or a richer temporal contract. Merely calling the survivor set “complete” does not add liveness.

Likewise, defining a greatest interior does not establish that it is inhabited. The empty set is always sound. A positive realization requires a witness, proved in corollary 3.3. This is an existence obligation, not an optional condition of Kernel governance: even the determinate judgment that a contract has no positive realization incurs the Kernel’s necessary work.

3 Greatestness, positive realization, and generated closure

Theorem 3.1 (Greatest admissible interior). *For the original contract of definition 2.1, define*

$$S = \bigcup \{U \subseteq X : U \subseteq \Phi(U)\}.$$

Then $S = \Phi(S)$. Every sound region is contained in S , and every fixed point of Φ is contained in S .

Proof. The operator Φ is monotone: $U \subseteq V$ implies $\Phi(U) \subseteq \Phi(V)$. If $x \in S$, some sound U contains x . Then $x \in B$ and every successor belongs to U , hence to S . Thus $S \subseteq \Phi(S)$. Monotonicity gives $\Phi(S) \subseteq \Phi(\Phi(S))$, so $\Phi(S)$ is itself sound and is included in the defining union S . This proves equality. The remaining statements follow directly from the union and the fact that a fixed point is sound. $\square$

This is the concrete form of [6, Theorem 6.3]; its order-theoretic background is [1]. The proof is included because the word “unique” below refers to the object just constructed.

Corollary 3.2 (Unique complete maximal region). *The set S is the unique inclusion-maximal sound region. A sound region I is complete for the contract, in the sense that it contains every sound region, if and only if $I = S$.*

Proof. Every sound region is included in the sound region S . A maximal one must therefore equal S . Conversely S is greatest and hence maximal. The completeness equivalence follows by applying completeness to S and soundness to I . $\square$

Greatestness is stronger than an unqualified appeal to maximality. It works here because unions of sound regions remain sound: for a point in a union, all successors lie in whichever sound member already contains that point. There is no hidden choice between incomparable maximal regions. This closure-of-unions fact must be reproved if the meaning of reuse is changed; Section 6 examines the important case of constructors with several arguments.

Corollary 3.3 (Positive realization). *The interior S is nonempty if and only if there exists a nonempty sound set U . More precisely, $x \in S$ if and only if some sound U contains x .*

Proof. Use the defining union in one direction and the sound set S in the other. $\square$

Nonempty immediate qualification does not suffice. Let $X = \{a, b\}$, $A = \{a\}$, and let the only edge be the warranted edge $a \rightarrow b$. Then $B = \{a\}$ but $S = \emptyset$. Conversely, an invariant loop $a \rightarrow a$ with $a \in B$ supplies a positive witness. The positive theorem must establish an invariant or equivalent global safety property, not simply exhibit an initially acceptable state.

Proposition 3.4 (Finite-depth construction and witnesses). *Put $U_0 = X$ and $U_{n+1} = \Phi(U_n)$. Then*

$$U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots, \quad S = \bigcap_{n < \omega} U_n.$$

A state belongs to U_n exactly when every state reachable from it by a path of length strictly less than n belongs to B . Consequently $x \notin S$ if and only if a finite path from x reaches a state outside B . No finite-branching hypothesis is required.

Proof. Descendingness follows from monotonicity and $U_1 \subseteq X$. The path characterization follows by induction: U_0 imposes no test, and $x \in U_{n+1}$ requires $x \in B$ and all successors in U_n . The operator preserves intersections of nonempty families, because

$$\text{Succ}(x) \subseteq \bigcap_i V_i \iff \forall i \text{ Succ}(x) \subseteq V_i.$$

Thus $U_\omega = \bigcap_n U_n$ satisfies $\Phi(U_\omega) = \bigcap_n U_{n+1} = U_\omega$. It is included in S by greatestness. Every sound set lies in every U_n , by induction, so $S \subseteq U_\omega$. The witness statement is the negation of the resulting universal finite-path condition. $\square$

For finite X , at most $|X|$ strict deletions occur before stabilization. For infinite X , this remains a mathematical characterization without providing an effective decision procedure for arbitrary input descriptions. The countable construction follows from this particular operator's intersection property; monotonicity alone on an arbitrary complete lattice would not justify the same countable bound.

The word “unique” also does not mean that Φ has only one fixed point. For $X = \{a, b\}$ with $B = X$ and only the self-loops $a \rightarrow a$, $b \rightarrow b$, every subset is a fixed point. Their unique greatest member is X . The complete interior is unique even when smaller self-maintaining regions exist.

Proposition 3.5 (When immediate qualification is already robust). *The equality $S = A$ holds if and only if $A \subseteq \Phi(A)$.*

Proof. Always $S \subseteq B \subseteq A$. If A is sound, greatestness gives $A \subseteq S$. If $S = A$, soundness of S gives soundness of A . $\square$

This is a substantive conservation condition: all immediately qualified candidates must have warranted edges leading back to immediate qualification. It is not secured by renaming A “admissible.”

3.1 Least generation and greatest admissibility

Definition 3.6 (Generated continuation closure). For a seed set $C \subseteq X$, let $\text{Cl}_R(C)$ be the states reachable by finite R -paths from some member of C , including paths of length zero.

Proposition 3.7 (The exact bridge from generation to completeness). *The set $\text{Cl}_R(C)$ is the least successor-closed superset of C . If $C \subseteq S$, then $\text{Cl}_R(C) \subseteq S$ and $\text{Cl}_R(C)$ is sound. Moreover,*

$$\text{Cl}_R(C) = S \iff C \subseteq S \text{ and every } s \in S \text{ is reachable from } C.$$

In particular, if $A \subseteq \Phi(A)$, then $S = A = \text{Cl}_R(A)$.

Proof. Closure follows by appending an edge; leastness follows by induction on path length in any successor-closed superset of C . Since S is successor-closed, $C \subseteq S$ implies $\text{Cl}_R(C) \subseteq S \subseteq B$. Together with successor closure, this proves soundness. Equality implies both stated conditions, including $C \subseteq S$ from zero-length paths. The reverse implication gives both set inclusions. The last assertion follows from proposition 3.5 and closure of the sound set A . $\square$

The bridge identifies what a corpus-generation claim owes. A least generated collection becomes the complete admissible interior only when its generators are sound and its coverage reaches every originally admissible case. Two isolated valid points with seed $\{a\}$ give $\text{Cl}_R(\{a\}) = \{a\} \subsetneq \{a, b\} = S$. Closure of a chosen corpus under its own operations cannot alone establish exhaustion of the original task. Under the soundness and coverage conditions, the least generated closure is exactly the greatest admissible interior.

4 Exact descent to a retained profile

The interior has so far been constructed on the original carrier. A representation can identify original states without identifying their continuation obligations. The next theorem gives an exact local test for when the safety operator, and consequently its greatest interior, descends through that identification. The test does not assume that the final standing answer already factors through the representation.

Fix the contract of definition 2.1, and let $q : X \rightarrow Q$ be a surjection. A subset of X is *q-saturated* when it has the form $q^{-1}(V)$ for some $V \subseteq Q$; equivalently it contains a whole fibre whenever it contains one point of that fibre. For $x \in X$ write

$$C_q(x) = q[\text{Succ}(x)] \subseteq Q.$$

As before, B combines immediate qualification with the warrants on every outgoing declared edge. Thus

$$x \in \Phi(U) \iff x \in B \text{ and } \text{Succ}(x) \subseteq U.$$

Theorem 4.1 (Exact one-step quotient criterion). *The following conditions are equivalent.*

(i) *There is an operator $\bar{\Phi} : \mathcal{P}(Q) \rightarrow \mathcal{P}(Q)$ such that*

$$\Phi(q^{-1}(V)) = q^{-1}(\bar{\Phi}(V)) \quad (V \subseteq Q). \quad (4.1)$$

(ii) *The operator Φ sends every q-saturated subset to a q-saturated subset.*

(iii) *For all $x, x' \in X$ with $q(x) = q(x')$,*

$$x \in B \iff x' \in B, \quad (4.2)$$

$$x \in B \implies C_q(x) = C_q(x'). \quad (4.3)$$

When these conditions hold, $\bar{\Phi}$ is unique. Put $\bar{B} = q[B]$ and define

$$\bar{C}(z) = \begin{cases} C_q(x), & z \in \bar{B} \text{ and } q(x) = z, \\ \emptyset, & z \notin \bar{B}. \end{cases}$$

This is well-defined, and the unique operator is

$$\bar{\Phi}(V) = \{z \in \bar{B} : \bar{C}(z) \subseteq V\}. \quad (4.4)$$

Proof. Condition (i) immediately implies (ii). Assume (ii). Since $X = q^{-1}(Q)$ and $\Phi(X) = B$, the set B is saturated; this gives (4.2). At a pair of locally valid states with $q(x) = q(x')$, saturation of $\Phi(q^{-1}(V))$ gives

$$C_q(x) \subseteq V \iff C_q(x') \subseteq V \quad (V \subseteq Q).$$

Taking $V = C_q(x)$ gives $C_q(x') \subseteq C_q(x)$, and taking $V = C_q(x')$ gives the reverse inclusion. Thus (iii) holds.

Assume (iii). Condition (4.2) makes $x \in B$ equivalent to $q(x) \in \bar{B}$, and (4.3) makes the displayed definition of $\bar{C}$ independent of the preimage used. No representatives are selected: the specified set has the same value at every preimage. For any $x \in X$,

$$\begin{aligned} x \in \Phi(q^{-1}(V)) &\iff x \in B \text{ and } q[\text{Succ}(x)] \subseteq V \\ &\iff q(x) \in \bar{B} \text{ and } \bar{C}(q(x)) \subseteq V \\ &\iff q(x) \in \bar{\Phi}(V). \end{aligned}$$

This proves (i) with (4.4). Finally, inverse image along a surjection is injective as a map between powersets: if $q^{-1}(V) = q^{-1}(V')$, every $z \in Q$ has a preimage witnessing $z \in V$ iff $z \in V'$. Hence (4.1) determines $\bar{\Phi}(V)$ uniquely for every V . $\square$

The qualification in (4.3) matters. At $x \notin B$, the conjunction defining $\Phi(U)$ is false for every U . Successor differences at such a source therefore impose no further constraint on this operator. If a separate original question asks about those successors or about the failed warrant, its answer must still be preserved by a representation claiming completeness for that question. The theorem does not erase such questions; it characterizes exactly the operator displayed above.

4.1 Greatestness commutes with the quotient

The relation $z \rightsquigarrow z'$ iff $z' \in \bar{C}(z)$, with local qualification $\bar{B}$ and every such edge warranted, realizes (4.4) as a contract of the same universal-safety form. Let $\bar{S}$ denote its greatest sound interior.

Theorem 4.2 (Exact descent of the complete interior). *Under the equivalent conditions of theorem 4.1, define*

$$U_0 = X, \quad U_{n+1} = \Phi(U_n), \quad V_0 = Q, \quad V_{n+1} = \bar{\Phi}(V_n).$$

Then, for every $n \geq 0$,

$$U_n = q^{-1}(V_n), \quad S = q^{-1}(\bar{S}), \quad q[S] = \bar{S}.$$

In particular, the greatest original interior is saturated and its attained image is the greatest quotient interior. For every $V \subseteq Q$,

$$q^{-1}(V) \subseteq \Phi(q^{-1}(V)) \iff V \subseteq \bar{\Phi}(V). \quad (4.5)$$

Moreover, if $U \subseteq X$ is sound, then both $q[U]$ and its full preimage $q^{-1}(q[U])$ are sound in their respective contracts. Thus saturating an independently sound region under this quotient preserves soundness.

Proof. The approximant identity holds at $n = 0$ by surjectivity and the codomain of q . If it holds at n , then theorem 4.1 gives

$$U_{n+1} = \Phi(q^{-1}(V_n)) = q^{-1}(\bar{\Phi}(V_n)) = q^{-1}(V_{n+1}).$$

By proposition 3.4, both universal-safety operators have greatest interiors given by the intersections of these descending sequences. Inverse image commutes with intersection, so

$$S = \bigcap_{n \geq 0} U_n = q^{-1} \left(\bigcap_{n \geq 0} V_n \right) = q^{-1}(\bar{S}).$$

Surjectivity gives $q[q^{-1}(\bar{S})] = \bar{S}$. For (4.5), substitute (4.1) and use the fact that inverse image along a surjection reflects as well as preserves inclusion.

Finally let U be sound and $z = q(x)$ for $x \in U$. Then $x \in B$ and $\text{Succ}(x) \subseteq U$, so $z \in \bar{B}$ and $\bar{C}(z) = q[\text{Succ}(x)] \subseteq q[U]$. Hence $q[U]$ is sound; (4.5) makes its full preimage sound as well. $\square$

This result applies to arbitrary branching. It neither requires a finite carrier nor bounds the number of continuations from a state. The countable intersection is justified by the particular universal-successor operator, as proved in proposition 3.4; the same formula is not asserted for an arbitrary monotone operator or for a requirement of eventual success.

The quotient identifies the complete answer without choosing a state in a positive fibre. If $z \in \bar{S}$ has several preimages, all of them belong to S . Their common qualification does not remove distinctions of original identity that the stated questions retain. Conversely, no point outside S can acquire standing merely because its notation is replaced by z : the proved preimage equality determines whether that replacement is faithful.

4.2 The exact witness to failed descent

Corollary 4.3 (A lost local distinction). *If q fails the criterion, there are x, x' in one q -fibre and a subset $V \subseteq Q$ such that exactly one of x, x' lies in $\Phi(q^{-1}(V))$. Consequently no operator on subsets of Q can satisfy (4.1). Such a witness can be obtained either with $V = Q$, or with $V = C_q(x)$ for a locally valid state x .*

Proof. If (4.2) fails, take $V = Q$, because $\Phi(X) = B$. Otherwise a failure of (4.3) supplies a locally valid equal-profile pair with different successor-profile sets. Exchange x and x' if necessary so that some element of $C_q(x')$ is not in $C_q(x)$, and take $V = C_q(x)$. Then x belongs to $\Phi(q^{-1}(V))$ and x' does not. An inverse image from Q cannot distinguish two members of one fibre. $\square$

4.3 What the criterion does, and does not, require

Three finite examples make the quantifiers inspectable. In each, all declared edges are warranted, so choosing $A = B$ realizes the stated data.

Equal final answers do not imply local descent. Let $X = \{a, b\}$, $B = \{a\}$, $\text{Succ}(a) = \{b\}$ and $\text{Succ}(b) = \emptyset$. Collapse a and b to one profile. The greatest interior is empty: b is immediately invalid, and the only continuation from a reaches b . Thus final interior membership is constant on the fibre. Nevertheless $\Phi(X) = \{a\}$ is not saturated. Descent of the final set alone is strictly weaker than descent of every one-step test.

Even full local validity does not determine successor profiles. Let $X = \{a, b, c\}$, $B = X$, and let the only successors be $a \rightarrow a$, $b \rightarrow c$, and $c \rightarrow c$. Set $q(a) = q(b) = u$ and $q(c) = v$, with $u \neq v$. Here $S = X$ and local validity is fibre-constant. But $C_q(a) = \{u\}$ and $C_q(b) = \{v\}$. The test $V = \{u\}$ distinguishes a from b after one step. Thus the second clause is not redundant even when the greatest interior already descends.

Invalid sources need not have matching successor profiles. Let $X = \{a, b, c\}$, $B = \{c\}$, and let the only successors be $a \rightarrow c$ and $c \rightarrow c$. Again set $q(a) = q(b) = u$ and $q(c) = v$. The successor-profile sets of a and b differ, but both states are locally invalid. The criterion holds, the

quotient has $\bar{B} = \{v\}$ and $\bar{C}(v) = \{v\}$, and $S = \{c\} = q^{-1}(\{v\})$. Requiring matching successors at invalid sources would be sufficient but unnecessarily strong.

These examples also separate necessity for *this exact theorem* from necessity for every weaker conclusion. A failed clause provides a witness to failed one-step descent. It need not provide a witness that final standing differs, because a later failure can make both states unsafe and a safe system can make both states safe.

Finally, $C_q(x)$ records successor profiles without their labels. That is sufficient and necessary for Φ , once all outgoing warrants have been included in B . It is not a full equivalence of label-sensitive observations or of trace languages. Such observations must be included in the independently specified complete profile and separately preserved. In particular, equality of nondeterministic trace answers must not be substituted for the proved equality of successor-profile sets.

5 Unique complete presentation and faithful transport

The quotient theorem now connects two kinds of uniqueness: the one greatest original interior and the one minimal complete information presentation. The connection requires the exact local preservation conditions just proved. Calling a representation “complete” does not, by itself, prove a continuation property that was omitted from its original questions.

5.1 The unique minimal complete presentation

Fix the independently specified family of original observations on X . Keep the totalized readers $p_j : X \rightarrow Y_j$ of Section 2, including the distinguished undefinedness answer where required. Restrict the codomain of their complete profile to its attained image:

$$P : X \rightarrow T, \quad P(x) = (p_j(x))_{j \in J}, \quad T = P[X].$$

Here T is the attained image, so P is surjective. The family of questions is fixed before competing representations are considered. The profile is not defined by assuming that a proposed robust-standing predicate is correct.

A *complete adequate presentation* is a surjection $h : X \rightarrow H$ through which every p_j factors. It is *minimal* when, in addition,

$$P(x) = P(x') \implies h(x) = h(x').$$

Minimality removes distinctions not needed for these original answers. It does not remove a required identity distinction, and it is not implied by adequacy alone. The elementary universal property below is established in the preceding foundation papers; its short proof fixes the correspondence needed for operator transport [6, 7, 8].

Theorem 5.1 (Unique minimal complete presentation). *Every complete adequate $h : X \rightarrow H$ has a unique surjection $u : H \rightarrow T$ such that $uh = P$. The map u is bijective if and only if h is minimal. For any two minimal complete presentations $h_i : X \rightarrow H_i$, there is a unique bijection*

$$v : H_1 \rightarrow H_2, \quad vh_1 = h_2.$$

It preserves every original answer through the respective readers.

Proof. If $h(x) = h(x')$, factorization of every original answer gives $P(x) = P(x')$. Hence $u(h(x)) = P(x)$ defines a function. Surjectivity of h makes it unique; surjectivity of P makes it surjective. If h is minimal, $u(h(x)) = u(h(x'))$ implies $h(x) = h(x')$, so u is injective. Conversely,

injectivity of u and $uh = P$ imply minimality. For two minimal presentations put $v = u_2^{-1}u_1$. This is a bijection with $vh_1 = h_2$. Every commuting map must have this value on the surjective image of h_1 , so it is unique. The original readers agree there because both return the original answer $p_j(x)$; surjectivity extends their agreement to all of H_1 . $\square$

The uniqueness is over the original carrier. Arbitrary bijections between abstract sets of equal cardinality have no such significance. Conversely an adequate presentation may retain a serial number or a derivation that the original questions do not use. It still decodes T , but its additional fibres prevent a claim that it is the unique minimal presentation.

Theorem 5.2 (Representation-independent operator and interior). *Let $h_i : X \rightarrow H_i$ be minimal complete presentations of the same original profile P , and let v be the unique bijection of theorem 5.1. The criterion of theorem 4.1 holds for one of P, h_1, h_2 if and only if it holds for all three. When it holds, let Φ_i and S_i be the descended operators and their greatest interiors. Then, for every $V \subseteq H_1$,*

$$\Phi_2(v[V]) = v[\Phi_1(V)], \quad S_2 = v[S_1], \quad S = h_i^{-1}(S_i). \quad (5.1)$$

The same bijection carries every finite-depth safety approximant of one presentation to the corresponding approximant of the other.

Proof. The three maps identify exactly the same pairs of original states. Their codomains are related by the bijections u_i and v . Thus constancy of B on fibres is the same condition for all three. For each x ,

$$h_2[\text{Succ}(x)] = v[h_1[\text{Succ}(x)]].$$

Since v is injective, equality of successor-profile sets for h_1 is equivalent to equality for h_2 ; the same argument applies to u_i and P . This proves equivalence of the criteria.

For the operator equation, take inverse images under h_2 and use $h_2 = vh_1$:

$$\begin{aligned} h_2^{-1}(\Phi_2(v[V])) &= \Phi(h_2^{-1}(v[V])) \\ &= \Phi(h_1^{-1}(V)) \\ &= h_1^{-1}(\Phi_1(V)) \\ &= h_2^{-1}(v[\Phi_1(V)]). \end{aligned}$$

Surjectivity of h_2 proves the first equality in (5.1). Since $v[H_1] = H_2$, induction gives the asserted equality of finite-depth approximants. A bijection preserves intersections; applying proposition 3.4 gives $S_2 = v[S_1]$. Finally theorem 4.2 gives $S = h_i^{-1}(S_i)$. $\square$

Changing a minimal complete presentation preserves both the descended operator and its complete greatest answer under the unique commuting identification. The operator must first be shown to descend. For example, a profile answering only present local validity is constant on a two-state system whose states are both valid, where one has no successors and the other has a self-loop. That profile completely answers its stated observation question, but $\Phi(\emptyset)$ distinguishes the states. It therefore does not support one-step descent. If the original task requires those continuation distinctions, its claim of observational coverage must be expanded accordingly.

5.2 Maps between distinct carriers

Faithful comparison can also use a map that is not a bijective recoding of one carrier. Let two independently specified contracts have carriers X, Y , effective local validity sets B_X, B_Y , successor maps $\text{Succ}_X, \text{Succ}_Y$, safety operators Φ_X, Φ_Y , and greatest interiors S_X, S_Y . An

interpretation $f : X \rightarrow Y$ is given; its existence or legitimacy is not inferred merely from matching notation.

Theorem 5.3 (Exact faithful transport). *For a function $f : X \rightarrow Y$, the following are equivalent.*

(i) For every $V \subseteq Y$,

$$\Phi_X(f^{-1}(V)) = f^{-1}(\Phi_Y(V)). \quad (5.2)$$

(ii) For every $x \in X$,

$$x \in B_X \iff f(x) \in B_Y, \quad (5.3)$$

$$x \in B_X \implies f[\text{Succ}_X(x)] = \text{Succ}_Y(f(x)). \quad (5.4)$$

The successor condition consists of two obligations at each locally valid source: every source successor maps to a target successor, and every target successor has a source successor mapping to it. The second obligation is successor lifting.

Under these conditions,

$$S_X = f^{-1}(S_Y), \quad f[S_X] = S_Y \cap f[X]. \quad (5.5)$$

If f is surjective, $f[S_X] = S_Y$. For every $V \subseteq Y$ that is sound, its preimage is sound. When f is surjective the converse holds as well. Every sound source region also has a sound direct image.

Proof. Assume (i). Set $V = Y$. Then (5.2) gives $B_X = f^{-1}(B_Y)$, which is (5.3). At a valid x , the same equation for every V gives

$$f[\text{Succ}_X(x)] \subseteq V \iff \text{Succ}_Y(f(x)) \subseteq V.$$

Taking each of these two sets in turn as V gives equality, proving (ii). Conversely, substitute (5.3) and (5.4) into the defining membership tests for the two sides of (5.2). At invalid x both tests fail; at valid x their successor conditions are equal. This proves (i).

Let $U_0 = X$, $U_{n+1} = \Phi_X(U_n)$ and $V_0 = Y$, $V_{n+1} = \Phi_Y(V_n)$. The equality $U_n = f^{-1}(V_n)$ holds at $n = 0$ and follows at $n + 1$ from (i). Intersecting and applying proposition 3.4 proves $S_X = f^{-1}(S_Y)$. Taking images gives the second equality in (5.5); surjectivity yields the further conclusion. If $V \subseteq \Phi_Y(V)$, taking preimages and using (i) proves soundness of $f^{-1}(V)$. If f is surjective, inverse image reflects inclusion, so the implication reverses.

If U is sound and $y = f(x)$ for $x \in U$, then $y \in B_Y$ and $\text{Succ}_Y(y) = f[\text{Succ}_X(x)] \subseteq f[U]$. Hence $f[U]$ is sound. This last assertion uses successor lifting as well as source soundness. $\square$

Surjectivity is a coverage requirement, not a requirement for the pointwise equivalence in (5.5). A faithful comparison may cover only part of the target. It then proves the target answer exactly at the interpreted source points; it does not establish that every target state has a source realization.

The theorem concerns the specified safety contracts. A stronger claim that f preserves every original observation must also prove the corresponding answer equations. If labels, numerical readings, original identities, or failure explanations matter, their readers and interpretation belong in those equations. Aggregating warrants into B proves the standing theorem; it does not automatically prove a label-sensitive equivalence of the systems.

5.3 Why forward matching alone is insufficient

The lifting clause cannot be suppressed. Let $X = \{s\}$ with $B_X = X$ and the warranted self-loop $s \rightarrow s$. Let $Y = \{t, b\}$ with $B_Y = \{t\}$ and warranted edges $t \rightarrow t$ and $t \rightarrow b$, and define $f(s) = t$. Local validity agrees, and the source edge maps to a target edge. But the extra target successor b has no lift. Thus $S_X = \{s\}$ while $S_Y = \emptyset$, contradicting the desired standing transfer. All pointwise and edge information actually checked in the forward direction was correct; it was incomplete.

Conversely, omit forward matching. Let $X = \{s, b\}$ with $B_X = \{s\}$ and its only edge $s \rightarrow b$. Let $Y = \{t, c\}$ with $B_Y = \{t\}$ and no edges, and put $f(s) = t$, $f(b) = c$. Local validity agrees, and the target has no successors requiring lifts. Yet $S_X = \emptyset$ and $S_Y = \{t\}$. The missing forward condition hid a source failure. Finally, without local-validity equivalence, two edge-free singleton carriers with different local qualifications already refute exact standing transport. Each hypothesis therefore has a concrete one-step role; no assumption of final-interior agreement is doing its work.

Proposition 5.4 (Composition and identity). *Identity maps satisfy the conditions of theorem 5.3. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ satisfy them for the indicated contracts, then $gf : X \rightarrow Z$ does too. If both maps are surjective, the composite covers the full target interior. In particular successive exact quotient reductions yield the same descended safety operator as direct reduction by their composite surjection.*

Proof. The identity assertions are immediate. Local validity is equivalent through both maps. At a valid x , $f(x)$ is valid, and

$$(gf)[\text{Succ}_X(x)] = g[f[\text{Succ}_X(x)]] = g[\text{Succ}_Y(f(x))] = \text{Succ}_Z(g(f(x))).$$

This proves the composite criterion. Surjectivity is preserved by composition, so theorem 5.3 gives full target coverage in that case. For successive quotient maps q and r , the one-step equations compose to

$$\Phi_X((rq)^{-1}(V)) = (rq)^{-1}(\Phi_Z(V)).$$

Uniqueness of the descended operator in theorem 4.1 identifies Φ_Z with the direct quotient operator. $\square$

The last assertion is an operator statement, not necessarily a statement about a chosen relation at invalid quotient states. The normalized quotient in theorem 4.1 assigns no successors there; other assignments at those invalid sources induce the same safety operator. No stronger identity of such unused relation data is inferred.

These results make standing conservation a proved preservation statement. A correct evaluation, translation, or construction can establish a new instance of the original requirements through its actual warrants. It gains no independent authority to alter the already fixed requirements, to ignore an unlifted continuation, or to repair the same failed occurrence by a change of presentation. Neither the quotient bijection nor a faithful many-to-one map is a theorem that only one original occupant exists.

6 Several contracts, one comparison problem

Uniqueness for one contract does not authorize the unqualified combination of separately safe protocols. A state can be safe for either protocol when used alone and fail when their steps alternate. This section gives the exact joint construction and the conditions under which adding

obligations preserves an established interior. Every comparison below uses the same carrier with fixed interpretations. If two source presentations use different carriers, their maps into that common comparison problem, and the obligations transported by those maps, must first be supplied.

Let I be a set of independently specified contracts on X . Contract i has local qualification and edge warrants aggregated into $B_i \subseteq X$, and declared successors $\text{Succ}_i(x)$. Write

$$\Phi_i(U) = \{x \in B_i : \text{Succ}_i(x) \subseteq U\}, \quad S_i = \nu\Phi_i.$$

Here $\nu\Phi_i$ denotes the greatest sound interior already constructed: the union of all U satisfying $U \subseteq \Phi_i(U)$. Labels, partial domains, and required readiness remain part of each original contract; aggregation does not license dropping a failed warrant or an unavailable required use. Define

$$B_* = \bigcap_{i \in I} B_i, \quad \text{Succ}_*(x) = \bigcup_{i \in I} \text{Succ}_i(x), \quad \Phi_*(U) = \bigcap_{i \in I} \Phi_i(U). \quad (6.1)$$

Thus $\Phi_*(U) = \{x \in B_* : \text{Succ}_*(x) \subseteq U\}$. For an empty family, the intersections are X and the successor union is empty.

6.1 Joint greatestness and the mixed-path obstruction

Theorem 6.1 (Unique greatest jointly sound interior). *For any set-indexed family above, put*

$$\mathcal{C}_* = \{U \subseteq X : \text{for every } i \in I, U \subseteq \Phi_i(U)\}, \quad S_* = \bigcup \mathcal{C}_*.$$

Then $S_ = \Phi_*(S_*)$. It is the unique greatest jointly sound region, and every inclusion-maximal jointly sound region equals it. Its safety condition covers finite paths whose successive edges may belong to different contracts; there is no restriction to repetitions of a single contract.*

Proof. Each Φ_i , and therefore Φ_* , is monotone. If $x \in S_*$, some $U \in \mathcal{C}_*$ contains x . For each i , joint soundness gives $x \in B_i$ and $\text{Succ}_i(x) \subseteq U \subseteq S_*$. Hence $S_* \subseteq \Phi_*(S_*)$. Monotonicity now gives $\Phi_*(S_*) \subseteq \Phi_*(\Phi_*(S_*))$, so $\Phi_*(S_*)$ itself belongs to $\mathcal{C}_*$. Its inclusion in the union proves the reverse containment. The definition of the union contains every jointly sound region. Consequently an inclusion-maximal such region must be S_* , since S_* is a sound region containing it.

For the path statement, joint soundness keeps a state in the same region after every edge from any Succ_i . Induction on path length therefore preserves membership and all local obligations under arbitrary finite interleavings. Conversely, let H be the states all of whose finite paths using Succ_* stay in B_* . The path of length zero gives $H \subseteq B_*$. Prepending one edge shows $\text{Succ}_*(x) \subseteq H$ for every $x \in H$. Thus H is jointly sound and is contained in S_* . The forward implication applied to S_* gives the opposite inclusion. This is a safety statement; it does not supply an eventual-success conclusion. $\square$

Theorem 6.2 (Exact intersection criterion). *Let $T = \bigcap_{i \in I} S_i$. Then*

$$S_* \subseteq T.$$

Equality holds if and only if T is sound for every contract:

$$S_* = T \iff \text{for all } i \in I, T \subseteq \Phi_i(T). \quad (6.2)$$

Equivalently, every successor of a point in T , under every contract, must remain in T . No finiteness assumption on I , X , or the branching is needed.

Proof. Every jointly sound region is sound for contract i , so greatestness gives $S_* \subseteq S_i$ for each i . If T is jointly sound, greatestness for Φ_* gives $T \subseteq S_*$, proving equality. Conversely, equality makes T jointly sound by theorem 6.1. Since $T \subseteq S_i \subseteq B_i$, its local qualification is already secured; the remaining condition is exactly successor closure in T . $\square$

Corollary 6.2.1 (Exact witness to strict common-interior loss). *With $T = \bigcap_{i \in I} S_i$ as above,*

$$S_* \subsetneq T \iff \exists i \in I \exists x \in T \exists y \in \text{Succ}_i(x) \setminus T.$$

Thus the common interior is strictly smaller than the intersection of the separate interiors exactly when some declared continuation from that intersection leaves it.

Proof. By theorem 6.2, $S_* \subseteq T$, with equality exactly when T is sound for every contract. Since $T \subseteq S_i \subseteq B_i$ for every i , all local validity conditions already hold on T . Failure of joint soundness is therefore exactly the displayed successor witness. $\square$

A strict intersection. Take $X = \{a, b, z\}$ and $B_1 = B_2 = \{a, b\}$. The only edge of contract 1 is $a \rightarrow b$; the only edge of contract 2 is $b \rightarrow z$. All displayed edges have their required warrants, and neither contract demands an unavailable operation at a terminal state. Then

$$S_1 = \{a, b\}, \quad S_2 = \{a\}, \quad T = \{a\}, \quad S_* = \emptyset.$$

Indeed, b fails the joint contract because its second-contract successor is $z \notin B_*$. The first-contract edge then excludes a . The mixed path $a \rightarrow b \rightarrow z$ is the concrete witness. Separate safety of a under each unmixed repertoire does not prove joint safety. The edge $a \rightarrow b$ already witnesses corollary 6.2.1: $a \in T$ but $b \notin T$. The common interior remains unique even when this interaction makes it empty.

6.2 Strengthening and conservative extension

The direction of contract strengthening must be explicit. Adding local requirements narrows B ; adding declared continuations enlarges the successor family to be checked. A claim that fewer transitions strengthen this universal safety contract would reverse the relevant order.

Theorem 6.3 (Exact one-step order and interior comparison). *Let two contracts on the same interpreted carrier have aggregate data (B, Succ) and (B', Succ') , operators Φ, Φ' , and greatest interiors S, S' . The following conditions are equivalent:*

- (i) $\Phi'(U) \subseteq \Phi(U)$ for every $U \subseteq X$;
- (ii) $B' \subseteq B$ and, for every $x \in B'$, $\text{Succ}(x) \subseteq \text{Succ}'(x)$.

Under these equivalent conditions, $S' \subseteq S$. In particular, $B' \subseteq B$ together with pointwise successor inclusion on all of X is sufficient.

Proof. Assume (i). At $U = X$, the operators return B' and B , giving $B' \subseteq B$. If $x \in B'$, take $U = \text{Succ}'(x)$. Then $x \in \Phi'(U) \subseteq \Phi(U)$, so $\text{Succ}(x) \subseteq \text{Succ}'(x)$. Conversely, if (ii) holds and $x \in \Phi'(U)$, then $x \in B' \subseteq B$ and $\text{Succ}(x) \subseteq \text{Succ}'(x) \subseteq U$. Thus $x \in \Phi(U)$. Finally $S' = \Phi'(S') \subseteq \Phi(S')$, so S' is sound for the first contract. Its greatest interior contains S' . $\square$

This equivalence concerns the complete one-step operators. It is not a necessary test for the mere inclusion $S' \subseteq S$: differences away from the greatest interiors may be irrelevant to that inclusion. For example, on $X = \{a, b\}$ take $B = B' = X$, with the sole original edge $a \rightarrow b$ and

no primed edges. Both greatest interiors are X , although (ii) fails at a . Nor does the theorem compare two differently interpreted original questions merely because their states have been given the same names.

Theorem 6.4 (Exact criterion for conservative repertoire extension). *Retain a contract Φ and add a set-indexed family of obligations (B_j^+, Succ_j^+) . Set*

$$\Psi_j(U) = \{x \in B_j^+ : \text{Succ}_j^+(x) \subseteq U\}, \quad \Phi^+(U) = \Phi(U) \cap \bigcap_j \Psi_j(U),$$

and let S^+ be the greatest interior of Φ^+ . Then $S^+ \subseteq S$, and

$$S^+ = S \iff \text{for every } j, \\ S \subseteq B_j^+ \text{ and } \bigcup_{x \in S} \text{Succ}_j^+(x) \subseteq S. \quad (6.3)$$

Thus extension preserves the complete interior exactly when all added local obligations and continuation obligations hold throughout that interior.

Proof. Since $\Phi^+(U) \subseteq \Phi(U)$ for all U , the comparison theorem gives $S^+ \subseteq S$. The displayed right-hand condition is precisely $S \subseteq \Psi_j(S)$ for every j . Together with $S = \Phi(S)$ it makes S sound for Φ^+ , giving $S \subseteq S^+$. Conversely, $S^+ = S$ and $S^+ = \Phi^+(S^+)$ imply every displayed obligation. $\square$

This is a preservation theorem, not a declaration that the added operations were already licensed. Their readiness, warrants, and results must be proved on S . Their behavior outside S can be arbitrary without changing the interior, provided the old contract and interpretation are retained. If extensions remove old requirements or alter their meaning, the intersection formula does not describe them and this theorem does not apply.

6.3 Many-input constructors and independently fixed contexts

The preceding results also cover operations with several arguments, provided their permitted contexts are specified independently of the region under test. Suppose, for clarity, that a partial constructor has type

$$c : X^n \multimap X$$

for a finite positive arity n . For each position r , fix the full original family Γ_r of partner tuples and any associated original context conditions. A partner tuple $\gamma \in \Gamma_r$ supplies all arguments except the r th and induces a partial one-input use

$$c_{r,\gamma}(x) = c(\gamma_1, \dots, \gamma_{r-1}, x, \gamma_{r+1}, \dots, \gamma_n).$$

Include every such permitted use in the declared relation. Its actual domain remains part of the specification. If readiness of this use is required at x , that readiness is a local condition in B ; an absent required result is not turned into a vacuous success by omitting its edge. If the use is optional outside its original domain, only the declared applicable cases impose an edge. A nondeterministic constructor contributes all its declared possible results. Multiple sorts can be handled on their tagged disjoint union, with the prescribed input and output tags retained.

Proposition 6.5 (Fixed-context constructor reduction). *For these independently fixed context families, a region U satisfies every declared constructor obligation if and only if it satisfies their aggregated local conditions and is closed under every induced applicable use $c_{r,\gamma}$. Consequently their intersection has the unique greatest jointly sound interior of theorem 6.1. A finite evaluation*

with initial operands in this interior has every intermediate result in the interior whenever each constituent use, including its partner context, belongs to the declared family and satisfies its required readiness and warrant.

Proof. An induced edge is, by construction, exactly an applicable original constructor use with one input position exposed and its other arguments fixed. Thus universal closure under those edges and satisfaction of their aggregated local conditions are exactly the stated obligations; each implication simply substitutes the declared context. The operator has the form $\{x \in B : \text{Succ}(x) \subseteq U\}$, so theorem 6.1 applies. For a finite evaluation, order its constituent uses so that arguments are evaluated before they are used. Each argument of the first use is an initial operand in the interior. Expose any one of its input positions; the other arguments give a declared partner context, so closure puts the result in the interior. Inducting through the remaining uses proves the assertion. The stated context-membership condition is required at every step; it does not follow from the existence of a syntax tree alone. $\square$

Why endogenous partner tests require another argument. The condition “ $c(x, y) \in U$ whenever $x, y \in U$ ” lets the permitted partner family vary with the candidate U . This is a different condition from closure under all independently declared contexts. Its sound regions need not be closed under unions. For example, take $X = \{a, b, z\}$, local qualification $\{a, b\}$, and a total binary operation with

$$c(a, a) = a, \quad c(b, b) = b, \quad c(a, b) = c(b, a) = z,$$

and $c(x, y) = z$ if either input is z . The singletons $\{a\}$ and $\{b\}$ are distinct inclusion-maximal qualified subalgebras. Their union is not a qualified subalgebra, because a mixed pair produces z .

Under the independently declared partner family $\{a, b\}$, by contrast, each of a and b must tolerate the other partner. Neither does, and the greatest sound interior for that fixed-context contract is empty. Both calculations are correct for their respective original requirements. One must not replace the former question by the latter without justification, or invoke the present uniqueness theorem for an endogenous closure condition whose union property fails. For a restricted partner family the theorem remains valid for exactly that family; extending its result to all original contexts requires a coverage proof.

The coverage boundary. Joint soundness does not by itself show that a finite program language, a list of contexts, or a table of protocols covers every use of the original problem. The finite induction proves preservation for generated uses whose primitive obligations and interpretations have been established. Export to the original task additionally requires that its relevant uses are represented and that interpretation preserves their domains, answers, and warrants. An infinite family of declared contexts is allowed, but eventual success along infinite executions is not obtained from universal finite-path safety. None of these obligations demands that every allowed use be physically run: they are facts to be established about the declared operations. When those facts are supplied, theorems 6.1, 6.2 and 6.4 give the complete joint uniqueness and preservation conclusions.

7 Conservation, warranted construction, and fixed targets

The uniqueness theorem determines conservation of standing, the warrants for new constructions, and the conditions for faithful scope transfer. Each result concerns a specified answer at a

specified incidence. A new qualified object is certified by its own warrants, and an established implication can be used wherever its hypotheses hold.

Theorem 7.1 (Conservation of the complete standing answer). *Let $c : X \rightarrow \{0, 1\}$ be the characteristic function of the greatest interior S . A reader $g : X \rightarrow \{0, 1\}$ is sound and complete for robust qualification exactly when $g = c$. More explicitly, the conditions*

$$g(x) = 1 \Rightarrow x \in S, \quad x \in S \Rightarrow g(x) = 1$$

imply agreement at every original candidate. If two readers both satisfy these conditions, they agree without a further agreement assumption.

Proof. At $x \in S$, completeness forces $g(x) = 1 = c(x)$. At $x \notin S$, soundness rules out $g(x) = 1$, so $g(x) = 0 = c(x)$. The reverse direction is immediate. $\square$

The mathematical step is elementary; the load-bearing obligations are the reader's original meaning, soundness, completeness, and coverage of X . One cannot prove agreement by including an already chosen classification in what it means to be the same task. Here S was constructed from independently specified qualification and continuations. A certificate whose positive set is a proper sound subset of S can be useful. It cannot claim to be the complete classifier while omitting other safe cases.

Corollary 7.2 (Standing-preserving redescription). *Let q satisfy theorem 4.1, with descended standing reader $\bar{c}$. Then $c = \bar{c} \circ q$. Under the conditions of theorem 5.3, the two contracts satisfy $c_X = c_Y \circ f$. Under the conditions of theorem 5.2, their unique commuting presentation map preserves every original answer and the full greatest interior.*

Proof. Take characteristic functions in the corresponding preimage equalities. $\square$

A change violating one of these standing equalities is not a standing-preserving redescription of the same incidence. It either misdescribes that incidence or changes what is being asked or which object is being used. Within the requested standing-bearing use, a violated necessary qualification is a degenerate case of that use. Merely possessing a new string, coordinate system, or attractive score cannot reauthorize it.

7.1 What construction can establish

Proposition 7.3 (Warranted outputs and unsupported proxies). *Let $f : C \rightarrow X$ construct or interpret outputs from a source set. Qualification of every output is equivalent to $f[C] \subseteq S$. It is established, for example, by a proved sound region U with $f[C] \subseteq U$. For a proposed exact proxy $m : X \rightarrow M$, a complete robust-standing decoder $d : m[X] \rightarrow \{0, 1\}$ exists if and only if c is constant on every m -fibre.*

Proof. The first equivalence is the meaning of universal output qualification; the sufficient condition follows from $U \subseteq S$. For the second, $c = d \circ m$ implies fibre constancy. Conversely fibre constancy defines $d(m(x)) = c(x)$ independently of the chosen preimage. $\square$

This result prohibits *unsupported* generation of standing. It does not prohibit an admissible constructor. A proof that an algorithm always returns a member of an invariant sound region is already a target warrant. It need not be followed by a separate ritual of rediscovering the same qualification. A source label, however, cannot replace that implication when the target's obligations differ.

The same applies to numerical carriers. A measured value may decide an original qualification if the interpretation and criterion are proved. There is no blanket ban on metric or quantitative tests inside a fixed task. What is excluded is an additional authority that changes the answer while purporting to conserve the same qualification. If a proposed score merges a safe and an unsafe original candidate, no deterministic processing of that unchanged score can recover their different correct standing answers: their identical inputs force identical outputs. Extra information can repair the interface, but then the information base has changed.

7.2 No retrospective repair of a retained incidence

For a finite path $\pi = (x_0, a_0, x_1, \dots, a_{n-1}, x_n)$ in the original relation, define the all-stages record predicate

$$H(\pi) = 1 \iff (\forall i \leq n \ x_i \in A) \text{ and } (\forall i < n \ (x_i, a_i, x_{i+1}) \in W).$$

This records what occurred along this path. It is different from robust standing, which quantifies over all relevant paths from a state.

Proposition 7.4 (Retention and continuation). *If $H(\pi) = 0$, then $H(\pi \cdot \rho) = 0$ for every extension retaining π as its prefix. If $x_0 \in S$, every state on every declared finite path from x_0 lies in S , and its record predicate is positive. A later independently qualified endpoint does not alter a negative retained prefix.*

Proof. An extension retains each failed local check or unwarranted edge of its prefix, proving the first claim. Since $S \subseteq \Phi(S)$, every edge from a state in S is warranted and has its target in $S \subseteq A$. Induction on path length proves the second claim. The first claim still applies when the last endpoint of an extension is independently qualified. $\square$

The failed load trial in Section 9 can be followed by engineering repair and a new successful trial. That new occurrence has its own qualification. It does not turn the original failed load incidence into a successful one. Correcting mistaken evidence about an earlier event is different again: it revises the asserted record against the fixed original question. The proposition concerns an actual negative prefix retained as such, not an infallibility claim about every description of history. Its mathematical content is prefix retention; it does not posit a universal physical arrow of time.

7.3 Target fixation and scope transfer

Suppose an original identity reader is $r_X : X \rightarrow O$ and a proposed target representation has reader $r_Y : Y \rightarrow O$. To use $f : X \rightarrow Y$ as a description of the same original target, the claim includes

$$r_Y \circ f = r_X.$$

For other originally required answers there are corresponding commuting equations. If an equation fails at x , the value pair supplies a witness that this representation is not preserving that original answer. A map can legitimately describe a different object or a different question, but that is not evidence for the unchanged target claim.

Faithful scope transfer therefore has two tasks: prove the relevant original-answer equations, and prove the qualification/continuation conditions needed for standing. Theorem 5.3 proves the second task exactly for the safety operator. Its successor-lifting counterexample shows why a positive source record alone cannot certify a target with an additional bad continuation. A transfer is thereby warranted exactly through the target obligations and original-answer equations established for it.

8 Exhaustion by explicit witnesses

The exhaustion argument associates each false completeness claim with a witness in the original problem. The categories below follow from negating exact equations and set inclusions, with coverage stated for all originally relevant uses.

Theorem 8.1 (Interior surplus or safe omission). *For any proposed complete interior $I \subseteq X$, either $I = S$, or at least one of the following witnesses exists:*

- (i) *a surplus point $x \in I \setminus S$ and a finite declared path from x to a state outside B ;*
- (ii) *an omitted point $x \in S \setminus I$ contained in a sound region U .*

Both kinds may occur. If I is sound and $I \neq S$, an omission necessarily occurs and no surplus occurs. If I is inclusion-maximal among sound regions, neither kind occurs and $I = S$.

Proof. Unequal subsets have a point in their symmetric difference. A point of $I \setminus S$ has the finite-path witness of proposition 3.4. A point of $S \setminus I$ has the sound witness supplied by corollary 3.3. If I is sound, greatestness gives $I \subseteq S$. If it is maximal, corollary 3.2 gives equality. $\square$

A state outside B either fails an original local qualification or has an outgoing edge lacking its required warrant. Thus the surplus witness ends at a determinate original defect. It does not merely restate a verdict in different terminology. For omission, an independently proved invariant U often gives a shorter certificate than recomputing the whole interior. These witness statements are classical existence results; effective discovery depends on how the original sets and transitions are presented.

Theorem 8.2 (Information loss or representational surplus). *Let P be the complete original profile and let $h : X \rightarrow H$ be an encoding, restricted to its attained image. Exactly one of the following adequacy alternatives holds:*

- (i) *every original answer factors through h ;*
- (ii) *there are $x, x' \in X$ with $h(x) = h(x')$ but $P(x) \neq P(x')$.*

If (i) holds, the presentation is minimal complete exactly when there is no pair with $P(x) = P(x')$ and $h(x) \neq h(x')$. Such a pair witnesses surplus information, not necessarily unsafe qualification.

Proof. Factorization implies fibre constancy. If fibre constancy holds, define the decoder by $d(h(x)) = P(x)$; this is well-defined and gives (i). Its failure gives (ii), and the two cannot both hold. Under (i), the final equivalence is the definition of minimality, or the injectivity criterion in theorem 5.1. $\square$

Proposition 8.3 (Exhaustion of exact quotient failures). *A surjection $q : X \rightarrow Q$ fails to support exact one-step descent if and only if either local validity differs within a fibre or two locally valid members of one fibre have different successor-profile sets. In each case an explicit subset $V \subseteq Q$ distinguishes the two states under $\Phi(q^{-1}(V))$.*

Proof. This is the negation of the two necessary-and-sufficient conditions in theorem 4.1, with the separating test constructed in corollary 4.3. $\square$

The conclusion being tested must be stated exactly. A failed quotient condition disproves the claim of *all one-step tests* descending. It need not disprove a weaker claim that the final standing bit happens to descend. Section 4 gives examples in which all final answers coincide although

the operator does not descend. Likewise, failure to verify the hypotheses of a faithful-transport theorem is not by itself a counterexample to the final standing equation. To refute that equation requires a specific x with $x \in S_X$ and $f(x) \notin S_Y$, or conversely. Distinguishing failed hypotheses from false conclusions prevents an impossibility argument from becoming a rule that labels every unproved proposal impossible.

8.1 The scope of a genuinely exhaustive catalogue

Suppose a corpus lists candidate realizations C and claims that testing them exhausts the original problem. Three routes can justify that claim: every original candidate is represented directly; every candidate has a proved faithful representative in the catalogue; or an independently proved schema covers every original candidate. A finite table of examples supplies none of those routes merely by having a last row. The generated-closure bridge in proposition 3.7 gives an exact version for reachable states; the generated-use coverage theorems of [7] give a compositional version for interpreted observations and operations.

Once coverage is established, the impossibility conclusion is unconditional for that original task: an alternative purported complete interior cannot be both sound and different. The proof does not enumerate possible notations. It quantifies over every $I \subseteq X$ and uses greatestness; the interface theorem quantifies over every encoding with the stated carrier and required answers. Novel terminology cannot evade those quantifiers. New original requirements are handled by Section 6, where their effect on the interior is computed rather than dismissed.

8.2 A practical defeat ledger

Claim challenged	Mathematical witness or obligation
Complete interior	An unsafe surplus path or an omitted point with a sound invariant.
Complete answer interface	Equal codes with different original required answers.
Minimal interface	Equal complete profiles with different retained codes.
Exact quotient operator	A same-fibre local-validity difference or a separating successor-profile subset.
Faithful standing transfer	A discordant source/target standing answer; failed local hypotheses instead delimit the theorem's applicability.
Same original target	A failed original-answer commutation equation.
Singleton occupant	Two qualified candidates with distinct original identities; a singleton proof needs existence and identity separation.
Full corpus coverage	An uncovered original use or candidate, or a gap in the asserted coverage proof.

The ledger specifies the exhaustive alternatives for realizations of the independently stated original problem. A false claim has the corresponding witness. An unsettled proof obligation remains open until that obligation is established or a counterexample is found.

9 Worked tests: protocol, representation, and occupant

9.1 Two safe bridge designs and an unsafe continuation

This finite protocol is an illustrative mathematical model, not an empirical bridge certification. Its original task is to sustain the declared load/unload cycle with locally sound states. All states

and continuations are fixed before proposing an interior. All listed edges have their stipulated protocol warrant; a warranted transition can still lead to a locally failed state.

State	Meaning	In A ?	Successor $T(x)$
a_0	Design A, ready	yes	a_1 (load)
a_1	Design A, loaded	yes	a_0 (unload)
b_0	Design B, ready	yes	b_1 (load)
b_1	Design B, loaded	yes	b_0 (unload)
w	Weak design, ready	yes	f (load)
f	Failed configuration	no	f (retain failure)

There are no other successors, and $B = A$. The calculation is

$$\begin{aligned}
U_0 &= \{a_0, a_1, b_0, b_1, w, f\}, \\
U_1 &= \{a_0, a_1, b_0, b_1, w\}, \\
U_2 &= \{a_0, a_1, b_0, b_1\} = U_3 = S.
\end{aligned}$$

The initially qualified state w is removed because its declared load reaches f . This is the explicit unsafe-path witness against $S \cup \{w\}$. By contrast, $I_A = \{a_0, a_1\}$ is sound but incomplete: $I_B = \{b_0, b_1\}$ independently certifies the omitted safe branch. Their union is sound and is the complete interior.

The two designs are distinct occupants within one complete region. They are not two rival complete interiors of the original six-state problem. Restricting the original candidates to one design creates a narrower task. Deleting the load edge out of w likewise changes the requested certification. If load availability is mandatory, that readiness condition belongs in A , so deleting the required operation yields local failure instead of vacuous success.

Repairing the weak structure and conducting a new trial can create a new successful incidence. The earlier path $w \rightarrow f$ remains a failed path. This is proposition 7.4 in familiar terms. A proof that the repair yields an invariant safe protocol supplies the new warrant under proposition 7.3; the theorem does not require a second test if the relevant target qualification has already been proved.

9.2 Twelve archival records, one faithful quotient

Let the raw record space be $D = X \times \{0, 1\}$, where the second coordinate is an archival-format tag that no original question consumes. Define $q(x, t) = x$. The original questions distinguish the six named protocol states and their required finite protocol answers. Define local validity by $(x, t) \in B_D$ iff $x \in A$, and put

$$\begin{aligned}
\text{Succ}_D(x, 0) &= \{(T(x), 0)\}, \\
\text{Succ}_D(x, 1) &= \{(T(x), 0), (T(x), 1)\}.
\end{aligned}$$

All these edges are warranted. The two records in a fibre can have different raw branching degrees. They nevertheless have the same successor profiles:

$$q[\text{Succ}_D(x, t)] = \{T(x)\}.$$

The local validity and successor-profile tests both hold. For every $V \subseteq X$,

$$\Phi_D(q^{-1}(V)) = \{(x, t) : x \in A, T(x) \in V\} = q^{-1}(\Phi_X(V)).$$

Consequently

$$S_D = \{a_0, a_1, b_0, b_1\} \times \{0, 1\}, \quad q[S_D] = S.$$

There are eight safe raw records and four safe original states, with one greatest interior at each level. The projection is not an isomorphism of the raw graphs: it is many-to-one and forgets different archival branching. Exact safety descent needs the proved profile condition, not an unnecessarily strong raw-graph identity.

Now encode the six original states by the distinct tokens 10, 11, 20, 21, 30, 40. The unique map carrying each original profile to its token conjugates the descended operator and carries the interior to $\{10, 11, 20, 21\}$. This is theorem 5.2. Keeping an additional unused bit is adequate but nonminimal; it is an implementation choice rather than a competing complete qualification.

For a failed interface, retain only $h(x, t) = \mathbf{1}_A(x)$. It merges $(a_0, 0)$ with $(w, 0)$. Both have code 1, but their successor-code sets are respectively $\{1\}$ and $\{0\}$. The subset $V = \{1\}$ witnesses failed one-step descent. More strongly, $a_0 \in S$ and $w \notin S$, so no decoder from this unchanged code can recover robust standing. The code also loses the required distinction between named protocol states. These are concrete failures of continuation preservation and original-answer completeness, respectively; no third truth value has appeared.

9.3 One occupant proved by independent constraints

The final example shows how a stronger domain conclusion can be obtained legitimately. Let original occupants be pairs $(u, v) \in \mathbb{R}_{\geq 0}^2$, with arbitrarily many notational encodings of each pair. The original compatibility requirements are

$$u + v = 10, \quad 2u + v = 14.$$

On the whole nonnegative plane define

$$j(u, v) = (u + v, 2u + v).$$

This readout is injective independently of the compatibility equations: equality of the two readout coordinates for two pairs gives $u = u'$ by subtraction, and then $v = v'$. Every compatible occupant has readout (10, 14). The pair (4, 6) is nonnegative and satisfies both equations, so it proves existence. Injectivity then proves that it is the only compatible original occupant.

The identity theorem in theorem 10.2 formalizes precisely this reasoning. It permits many representations of the one pair. The numerical answer came from the original equations, their solved existence witness, and injective reconstruction. Uniqueness of the complete qualification region ensures that faithful complete presentations cannot disagree about that region; it does not replace the calculation proving that the region contains only one identity.

For comparison, replace the equations by $u + v = 10$ alone. The readout $u + v$ is no longer injective on original pairs; (4, 6) and (5, 5) are different qualified occupants. The complete solution set is still uniquely determined. This is exactly the distinction that a downstream uniqueness argument must preserve.

10 The reference theorem and its corpus exports

Theorem 10.1 (Complete admissible realization for one original task). *Fix an independently interpreted original contract D and its original answer profile P . Then:*

- (i) *There is exactly one complete maximal sound interior S . It is inhabited exactly when there is a nonempty sound witness. Every proposed different complete interior has a surplus or omission witness of theorem 8.1.*
- (ii) *Minimal complete presentations of P are uniquely isomorphic by the map commuting with the original carrier. If the exact quotient criterion holds for P , all those presentations carry the same safety operator, every finite-depth approximation, and the same greatest interior under that unique identification.*
- (iii) *A faithful map satisfying theorem 5.3 preserves and reflects standing at every interpreted point. Surjectivity gives full target coverage; compositions preserve these properties.*
- (iv) *A set-indexed collection of contracts on the same carrier has one greatest jointly sound interior. It equals the intersection of the separate interiors exactly when that intersection is sound for every contract. Added obligations preserve an old interior exactly under theorem 6.4.*
- (v) *Same-task sound and complete standing readers agree. A change of notation, proxy, or finite catalogue cannot alter the complete original answer unless the corresponding original-answer, safety, or coverage obligation fails.*

Proof. Part (i) is theorems 3.1 and 8.1 and corollaries 3.2 and 3.3. Part (ii) is theorems 4.2, 5.1 and 5.2. Part (iii) is theorem 5.3 and proposition 5.4. Part (iv) is theorems 6.1, 6.2 and 6.4. Part (v) follows from theorems 7.1 and 8.2 and proposition 7.3, together with the fixed-original-task requirement and the coverage bridge in proposition 3.7. No part assumes agreement between rival standing readers or inserts S into the original data. $\square$

The theorem determines one complete maximal interior and its exact faithful realizations for the independently fixed problem. Soundness excludes unqualified cases, and completeness includes every qualified case. A relabelling preserving the complete structure realizes the same theorem. A different proposed complete structure has the surplus, omission, or representation witness specified above.

10.1 The additional bridge to one occupant

Theorem 10.2 (Exact occupant-uniqueness bridge). *Let Z be an original candidate space, $\nu : Z \rightarrow O$ its original identity map, and $C \subseteq Z$ the independently specified compatible family. Suppose $j : O \rightarrow Y$ is injective and the original readout is $r = j \circ \nu$. Then*

$$\nu[C] \text{ is a singleton} \iff C \neq \emptyset \text{ and } r \text{ is constant on } C.$$

If only constancy is proved, $\nu[C]$ has at most one element. These conclusions are preserved by a surjective interpretation of a second candidate family that commutes with original identity and readout.

Proof. If $\nu[C] = \{o\}$, some candidate realizes o and every candidate has readout $j(o)$, proving the right side. Conversely choose an exhibited $z_* \in C$. For every $z \in C$, constancy gives $j(\nu(z)) = j(\nu(z_*))$. Injectivity yields $\nu(z) = \nu(z_*)$, and nonemptiness gives a singleton rather than the empty set. Without existence, the same pairwise argument gives at most one. A

surjective interpretation preserving identity and readout makes the attained identity and readout images of the two families equal, so the conclusions transfer. $\square$

Application of the theorem requires three domain proofs: constancy from independent original requirements, injectivity for the identity distinction claimed, and a positive realization. The compatible family C is specified independently of the singleton conclusion. The two-equation example in Section 9 proves all three obligations on a larger original candidate space. A Boolean qualification bit alone cannot provide an identity-separating readout on a family containing several positive occupants.

10.2 Exact export table

Result available here	Application must establish	Conclusion transferred
Theorem 3.1 and corollary 3.2	Original candidates, qualification, warrants, and all claimed universal continuations.	One greatest complete sound interior.
Corollary 3.3	A nonempty invariant sound witness.	A positive realization, rather than merely a defined greatest set.
Proposition 3.7	Sound seeds and coverage of all safe original candidates by declared paths.	Least generated closure equals the complete interior.
Theorems 4.1 and 4.2	Surjective interface, fibre-constant local validity, equal successor classes at valid sources.	Exact safety descent, standing preservation and reflection, transported invariants.
Theorems 5.1 and 5.2	Same original complete profile, minimality, and the quotient criterion.	Unique commuting presentation map and conjugate safety operators.
Theorem 5.3 and proposition 5.4	Local validity equivalence, forward matching, successor lifting; original readers separately commute.	Exact standing transfer at interpreted points; full coverage if surjective.

Result available here	Application must establish	Conclusion transferred
Theorems 6.1 and 6.2 and corollary 6.2.1	Independently interpreted contracts on one carrier, with joint switching included.	One common interior; exact equality test and successor witness to strict loss from the intersection.
Theorem 6.4	Every added obligation holds throughout the old interior.	Conservative extension of its complete standing answer.
Proposition 7.4	Same retained incidence and prefix, same original local/warrant questions.	Later success does not erase an actual earlier failure.
Theorems 8.1 and 8.2 and proposition 8.3	The proposed answer, interface, or quotient and coverage of its original claim.	Explicit witnesses to every false equality of the specified kind.
Theorem 10.2	Nonempty compatible family, independently forced common readout, injective original-identity reconstruction.	Exactly one occupant identity, preserved through faithful covering representations.

The first row establishes the greatest complete sound interior. Physical laws, numerical values, and occupant identities require their domain constraints and existence proofs; the last row gives the exact bridge for occupant uniqueness. Each row specifies the object determined and the hypotheses required for its determination.

10.3 Three corpus applications and their domain conditions

The physical-clock manuscript distinguishes these levels: its Proposition 15.6 permits zero, one, or several canonical clock classes inside one admissible interior, while Definitions 15.7–15.8 distinguish closure, realization, and singularity [9]. The present theorem supports the unique complete comparison region once the original clock task and faithful interfaces have been established. Counting clock classes remains a separate question about occupants or quotient classes within it.

The charged-lepton scalar-ratio manuscript explicitly treats global interior uniqueness as background in Remark 8.1, then adds nonemptiness and at-most-one arguments for its scalar slot in Lemma 8.2 and Theorem 9.1 [10]. The export here identifies the general obligations behind such a step: coverage of the original compatible candidates, an independently fixed governing readout, a proved common value, and the identity or output separation required by the target claim. A function being single-valued at one fixed input does not itself prove that every compatible physical candidate has that input and that same governing interpretation.

The ensemble-embedding manuscript’s Theorem 16.3 and Corollary 16.4 use same-interior uniqueness and inherited comparison standing [11]. Its application requires the following condition: a proposed envelope must represent the same independently specified contract with the required coverage and faithful transport. Distinct safe members of an interior are not thereby identical complete frameworks. An embedding introducing additional target obligations is governed by the joint-contract and conservative-extension theorems.

These applications receive the foundation theorems through their stated hypotheses. Establishing the domain interpretation and discharging those hypotheses yields the corresponding local conclusion. The dependence runs from the construction and transport results proved here to the particular physical or mathematical application.

10.4 Conclusion

The complete admissible interior is determined by the original qualification and reuse contract. Its uniqueness follows from greatestness; its representation independence follows from exact descent; its positive realization follows from a sound witness; and its exhaustive defence follows from original surplus, omission, and information-loss witnesses. The same structure survives faithful comparison and composition. Several contracts have a unique common interior, with a precise condition governing whether their separately safe regions remain jointly safe.

Unus establishes one complete admissibility structure for an independently specified construction problem, together with its exact representations, faithful transports, and conditions for occupant uniqueness. The Kernel's necessary governance remains in force throughout. The theorem chain identifies the complete interior for the specified task and proves how that determination is conserved.

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